Test Series No.

1.5

MASTERY TEST SERIES

DURATION: 90 Minutes | **M. MARKS: 400**

DATE: 16/02/2026

ANSWER KEY

1. (3)	26. (3)	51. (1)	76. (3)
2. (1)	27. (4)	52. (4)	77. (2)
3. (1)	28. (2)	53. (3)	78. (1)
4. (3)	29. (4)	54. (2)	79. (2)
5. (3)	30. (1)	55. (3)	80. (4)
6. (3)	31. (3)	56. (2)	81. (3)
7. (1)	32. (3)	57. (2)	82. (3)
8. (2)	33. (1)	58. (3)	83. (3)
9. (2)	34. (2)	59. (3)	84. (2)
10. (2)	35. (2)	60. (3)	85. (2)
11. (3)	36. (1)	61. (4)	86. (4)
12. (1)	37. (4)	62. (2)	87. (2)
13. (2)	38. (1)	63. (3)	88. (2)
14. (3)	39. (3)	64. (4)	89. (2)
15. (1)	40. (4)	65. (3)	90. (3)
16. (1)	41. (3)	66. (2)	91. (4)
17. (2)	42. (1)	67. (1)	92. (3)
18. (3)	43. (2)	68. (1)	93. (1)
19. (4)	44. (3)	69. (3)	94. (1)
20. (3)	45. (1)	70. (1)	95. (1)
21. (2)	46. (3)	71. (3)	96. (3)
22. (4)	47. (1)	72. (2)	97. (2)
23. (2)	48. (4)	73. (3)	98. (4)
24. (4)	49. (1)	74. (1)	99. (3)
25. (1)	50. (3)	75. (4)	100. (2)

[Contd...

1. (3)

In case of EM wave $E = E_0 \sin(kx - \omega t)$, speed is equal to $\frac{\omega}{k}$ which is independent of wavelength.

(NCERT Ed. 2025-26, 12th, Page No. 207)

2. (1)

Magnetic susceptibility has large positive value for ferromagnetic material.

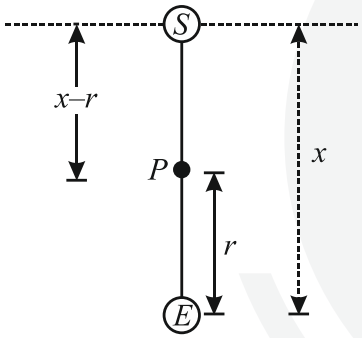
(NCERT Ed. 2025-26, 12th, Page No. 147)

3. (1)

Given, mass of the sun $M_s = 2 \times 10^{30}$ kg

Mass of the earth $M_e = 6 \times 10^{24}$ kg, orbital radius $= 1.5 \times 10^{11}$ m

Let, P be a point at a distance r from the Earth's centre, where gravitational force due to Sun and Earth are equal and opposite and hence, gravitational force on the rocket is zero.



$$\frac{GM_s m}{(x-r)^2} = \frac{GM_e m}{r^2}$$

$$\frac{M_s}{(x-r)^2} = \frac{M_e}{r^2}$$

$$\frac{2 \times 10^{30}}{(x-r)^2} = \frac{6 \times 10^{24}}{r^2}$$

$$\frac{(x-r)^2}{r^2} = \frac{2 \times 10^{30}}{6 \times 10^{24}} = \frac{1}{3} \times 10^6$$

$$\frac{x-r}{r} = \frac{10^3}{\sqrt{3}}$$

$$\frac{1.5 \times 10^{11}}{r} = \frac{10^3}{\sqrt{3}} \quad (\because x - r \approx 1.5 \times 10^{11} \text{ m})$$

$$r = 1.5 \times \sqrt{3} \times 10^8$$

$$r = 2.598 \times 10^8 \text{ m}$$

$r \approx 2.6 \times 10^8$ m (from the earth)

(NCERT Ed. 2025-26, 11th, Page No. 130)

4. (3)

As magnet moves towards right, magnetic flux linked to loop B will increase and for loop A it will decrease, hence induced current will be in opposite direction for loop A and B.

(NCERT Ed. 2025-26, 12th, Page No. 160)

5. (3)

In S.H.M velocities of a particle at distance x_1 and x_2 from mean position.

$v_1 = \omega\sqrt{A^2 - x_1^2}$ ....(i)

Similarly,

$v_2 = \omega\sqrt{A^2 - x_2^2}$ ....(ii)

Subtracting (ii) from equation (i), we get—

$$v_1^2 - v_2^2 = \omega^2(A^2 - x_1^2) - \omega^2(A^2 - x_2^2)$$

$$v_1^2 - v_2^2 = \omega^2(x_2^2 - x_1^2)$$

$$\omega^2 = \frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}$$

$$\omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}$$

$$\Rightarrow \omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}$$

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6. (3)

Given that, radius of curvature of convex mirror (C) $= 1.6$ m

Focus of convex mirror $(f) = \frac{1.6}{2} = +0.8$ m

Distance of object $(u) = -1$ m

From mirror formula—

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{0.8} = \frac{1}{-1} + \frac{1}{v}$$

$$\frac{1}{v} = 1 + \frac{1}{0.8}$$

$$\frac{1}{v} = \frac{1.8}{0.8}$$

$$v = \frac{0.8}{1.8}$$

$$v = \frac{8}{18}$$

$v = \frac{4}{9}$ m behind the mirror.

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[Contd...

7. (1)

In a plane electromagnetic wave, the electric field $(\vec{E})$, magnetic field $(\vec{B})$, and the direction of propagation $(\vec{k})$ are mutually perpendicular. The direction of propagation is given by $\vec{k} = \vec{E} \times \vec{B}$. Given that the wave travels in the Z-direction, we need $\vec{E} \times \vec{B}$ to be in the Z-direction.

For option (1): $\hat{i} \times \hat{j} = \hat{k}$ (Z-direction). This is correct.

(NCERT Ed. 2025-26, 12th, Page No. 206)

8. (2)

$X_L = \text{angular frequency} \times \text{Inductance}$

$X_L = 1000 \times 0.9 = 900 \text{ ohms}$

$X_C = 1/(\text{angular frequency} \times \text{Capacitance})$

$X_C = 1/(1000 \times 0.000002) = 500\ \Omega$

$X = X_L - X_C$

$X = 900 - 500 = 400 \text{ ohms}$

Now resistance $R = 300\ \Omega$ and Net Reactance $X = 400\,\Omega$.

$Z = 500 \text{ ohms}$.

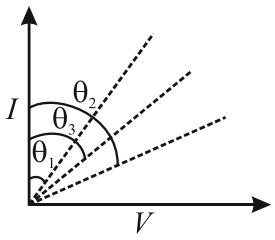
(NCERT Ed. 2025-26, 12th, Page No. 187)

9. (2)

By Ohm's law,

$V \propto I$

$V = IR$



$\frac{V}{I} = \tan\theta = R \quad (\tan\theta \uparrow, \theta \uparrow)$

The graph between current and voltage is

$\theta_1 < \theta_3 < \theta_2$

$\tan\theta_1 < \tan\theta_3 < \tan\theta_2$

$R_1 < R_3 < R_2$

(NCERT Ed. 2025-26, 12th, Page No. 83)

10. (2)

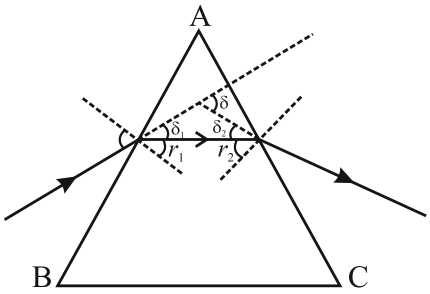
for minimum deviation, $\delta_m = A$

We know that,

$\delta + A = i + e \ (\because i = e)$

$2A = 2i$

$A = i$



Hence, $r_1 = r_2 = \frac{A}{2}$

$i = A = 90^\circ$

from Snell's law —

$\mu_0 \sin i = \mu \sin r_1$

$\sin A = \mu \sin(A/2)$

$\mu = \frac{2\sin(A/2)\cdot\cos(A/2)}{\sin(A/2)}$

or $\mu = 2\cos(A/2)$

To find the range of μ, we look at the possible values of the prism angle A:

For a very thin prism, the angle A approaches 0°.

As A approaches 0, $\cos\left(\frac{A}{2}\right)$ approaches $\cos(0)$, which is 1.

So, $\mu = 2 \times 1 = 2$.

For light to emerge from the prism at minimum deviation, the angle of refraction inside the prism (which is $\frac{A}{2}$) must not exceed the critical angle (C).

The condition for emergence is $\frac{A}{2}$ is less than or equal to C.

At the limit where $\frac{A}{2} = C$:

$\sin\left(\frac{A}{2}\right) = \sin(C) = \frac{1}{\mu}$

Substitute $\mu = 2\cos\left(\frac{A}{2}\right)$ into this:

$\sin\left(\frac{A}{2}\right) = \frac{1}{2\cos\left(\frac{A}{2}\right)}$

[Contd...

$$2\sin\left(\frac{A}{2}\right)\cos\left(\frac{A}{2}\right)=1$$

$\sin(A)=1$

$A = 90°$.

For $A = 90°$,

$$\mu = 2\cos\left(\frac{90°}{2}\right) = 2\cos(45°) = 2\times\left(\frac{1}{\sqrt{2}}\right) = \sqrt{2}\,.$$

Therefore, the refractive index μ must lie between $\sqrt{2}$ and 2.

(NCERT Ed. 2025-26, 12th, Page No. 240)

11. (3)

Given, Radius of air bubble $(r) = 0.01$ m

Terminal velocity $(v_T) = 0.25\times10^{-2}$ m/s

Density of liquid $(\rho) = 1.5\times10^{3}$ kg/m^3

Density of air $(\sigma_{air}) = 0$, $g = 9.8$ m/s^2, $\eta = ?$

We know that, terminal velocity

$$(v_T) = \frac{2r^2(\rho-\sigma_{air})g}{9\eta}$$

or $$\eta = \frac{2r^2(\rho-\sigma_{air})g}{9v_T}$$

$$\eta = \frac{2}{9}\times\frac{(0.01)^2(1.5\times10^3-0)\times9.8}{0.25\times10^{-2}} = 130.667$$

$\eta = 130.667 \approx 130$ Pa-s

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12. (1)

(A) Microwaves: Used in radar systems, microwave ovens. So, (A) matches with (III).

(B) Ultraviolet rays: Used in sterilization of medical equipment, water purification. So, (B) matches with (IV).

(C) Infrared waves: Used in remote controls. So, (C) matches with (I).

(D) X-rays: Used for medical imaging (e.g., bone fractures), security scanners at airports. So, (D) matches with (II).

(NCERT Ed. 2025-26, 12th, Page No. 211)

13. (2)

$|\vec{\tau}| = MB\sin\theta$

$U = -MB\cos\theta$

At θ = 90°, torque is maximum and potential energy is zero.

(NCERT Ed. 2025-26, 12th, Page No. 139)

14. (3)

RMS Current = Peak Current / $\sqrt{2}$

Peak Current = $\sqrt{2}\times$ RMS Current

Ratio = $(\sqrt{2}\times\text{RMS})$ / RMS

Ratio = $\sqrt{2}$

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15. (1)

Given :

radius $(r) = 0.4$ m, Mass $(m) = 1$kg

Angular acceleration $(\alpha) = 10$ rad/s^2

Tangential force $(F) = ?$

Moment of inertia of disc $I = \frac{mr^2}{2}$

$$I = \frac{1\times(0.4)^2}{2}$$

$I = 0.08$ kg m^2

$\because$ Torque, $\tau = I\alpha$

$\tau = 0.08\times10 = 0.8$ N m

$$\tau = Fr \Rightarrow F = \frac{\tau}{r} = \frac{0.8}{0.4} = 2 \text{ N}$$

(NCERT Ed. 2025-26, 11th, Page No. 120)

16. (1)

$$P = \frac{1}{f} = \left(\frac{\mu_l}{\mu_m}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$\mu_l = \mu_m$

$$P = \frac{1}{f} = (1-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$P = 0$

(NCERT Ed. 2025-26, 12th, Page No. 234)

17. (2)

The speed is given in km/h. Always convert to m/s first.

Speed $(v) = 72\times(5/18) = 20$ m/s.

Length $(l) = 50$ cm $= 0.5$ m.

Use the formula: EMF $= Bvl$

$10 = B\times20\times0.5$

$10 = B\times10$

$B = 1$ T

(NCERT Ed. 2025-26, 12th, Page No. 162)

18. (3)

$$V_A - iR + \varepsilon - L\frac{di}{dt} = V_B$$

$$V_A - (2\times 3) + 7 - (12\times 10^{-3})\frac{di}{dt} = V_B$$

$$V_A - V_B = -1 + (12\times 10^{-3})\frac{di}{dt}$$

$$1 = 12\times 10^{-3}\frac{di}{dt}$$

$$\frac{di}{dt} = \frac{1000}{12}$$

$$= \frac{250}{3} \text{ A/s}$$

(NCERT Ed. 2025-26, 12th, Page No. 168)

19. (4)

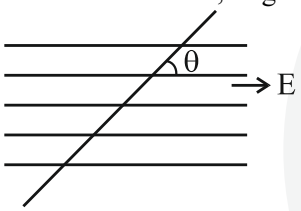
Incident and emergent rays will be parallel therefore angle between them will be zero.

(NCERT Ed. 2025-26, 12th, Page No. 239)

20. (3)

Let electric dipole moment = p

Electric field = E, angle = θ



Since, we know that,

$U = -pE\cos\theta$

If $\theta = 0$ (for minimum)

$U = -pE\cos 0°$ $[\because \cos 0^\circ = 1]$

$U = -pE$

Dipole for stable equilibrium—

$\theta = 90°$

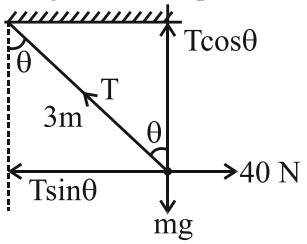
$U = -pE\cos\theta = -pE\cos 90° = 0$

(NCERT Ed. 2025-26, 12th, Page No. 27)

21. (2)

Given, mass $(m) = 8$ kg $= 80$ N, force $F = 40$ N

Length of the rope $= 3$ m



Let θ angle make by the rope with vertical for equilibrium—

From the figure,

$\sum F_x = 0$

$T\sin\theta = 40$ .....(i)

And $\sum F_y = 0$

$T\cos\theta = mg = 80$ .....(ii)

From equation (i) / (ii), we get—

$$\frac{T\sin\theta}{T\cos\theta} = \frac{40}{80} \Rightarrow \tan\theta = \frac{40}{80} \Rightarrow \theta = \tan^{-1}(1/2)$$

(NCERT Ed. 2025-26, 11th, Page No. 58)

22. (4)

Magnetic field due to current carrying long wire is,

$$B = \frac{\mu_0 I}{2\pi r} \Rightarrow B \propto \frac{1}{r}$$

It is clear from the equation that the relationship between B and r is rectangular hyperbolic.

(NCERT Ed. 2025-26, 12th, Page No. 120)

23. (2)

Given,

Length & area are equal i.e.

$AB = BC = CD = L$

$(A)_{AB} = (A)_{BC} = (A)_{CD} = A$

And thermal conductivity,

$(K)_{AB} = 2K, (K)_{BC} = 4K, (K)_{CD} = 3K$

Equivalent thermal conductivity of the given series,

$$\frac{3L}{K_{eq}A} = \frac{L}{2KA} + \frac{L}{4KA} + \frac{L}{3KA}$$

$$K_{eq} = \frac{36K}{13}$$

$\because (Q)_{AD} = (Q)_{AB}$

$$K_{eq}\cdot A\frac{(200-18)}{3L} = \frac{2K\cdot A(200-\theta)}{L}$$

$$\frac{36K}{13}\times\frac{(200-18)}{3} = 2(200-\theta)$$

$$\frac{6}{13}\times 182 = 200-\theta$$

$\theta = 116°$C

So, the temperature at B, $\theta = 116°$C

(NCERT Ed. 2025-26, 11th, Page No. 216)

24. (4)

$$\omega = \frac{50}{\pi}\times 2\pi = 100 \text{ rad/s}$$

$$e = \frac{1}{2}B\omega L^2 = \frac{1}{2}\times(0.5)(100)(2)^2$$

$= 100$ volt

(NCERT Ed. 2025-26, 12th, Page No. 164)

25. (1)

$$\frac{E_{rms}}{B_{rms}} = c$$

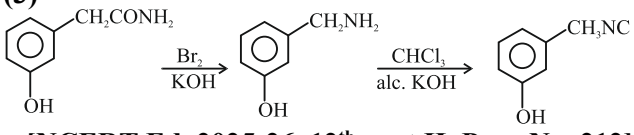
$$B_{rms} = \frac{6}{3\times 10^8} = 2\times 10^{-8}\text{T}$$

$$B_0 = B_{rms}\sqrt{2} = 2\times 10^{-8}\times\sqrt{2} = 2\sqrt{2}\times 10^{-8}\text{T}$$

(NCERT Ed. 2025-26, 12th, Page No. 207)

[Contd...

26. (3)



[NCERT Ed. 2025-26, 12th part II, Page No. 212]

27. (4)

Apply, $\Delta T_f = K_f m$

$0.48 = K_f \times m$ …(1)

$\Delta T_b = i K_b m$

$0.26 = i K_b m$ …(2)

On dividing eq. (1) and (2), we get

$$\frac{0.48}{0.26} = \frac{1}{i} \times 3.7 \qquad \left(\because \frac{K_f}{K_b} = 3.7\right)$$

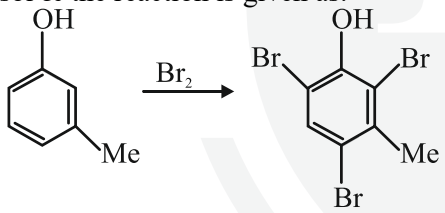
$$i = \frac{3.7}{1.84} = 2$$

Since, van't hoff factor, i > 1, it's a case of dissociation.

[NCERT Ed. 2025-26, 12th part I, Page No. 18]

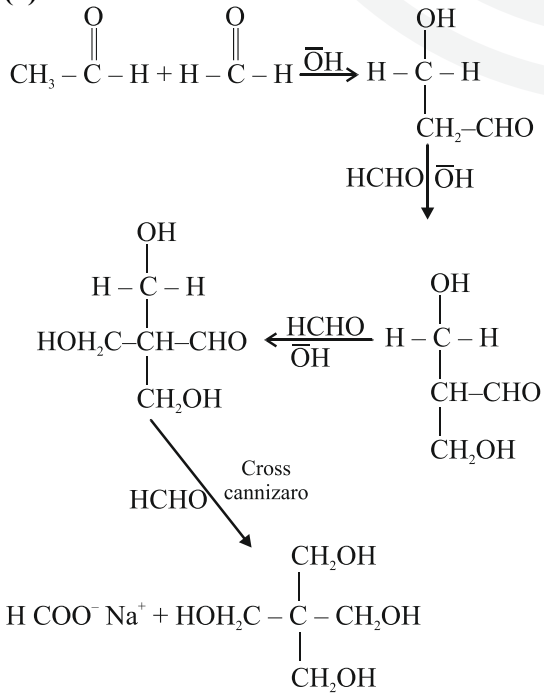
28. (2)

C_7H_8 is soluble in NaOH but not in $NaHCO_3$. So, —OH group is directly attached to the benzene ring. C_7H_8O reacts with Br_2 to give CH_5OBr_3. So, two ortho and one para position should be vacant w.r.t.—OH group. Hence, $C_7H_8O(X)$ must be m-cresol & the reaction is given as:



[NCERT Ed. 2025-26, 12th part II, Page No. 212]

29. (4)



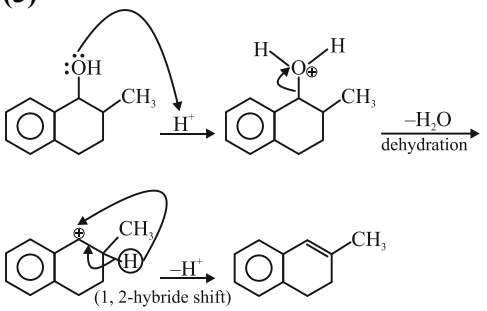
[NCERT Ed. 2025-26, 12th part II, Page No. 242]

30. (1)

Maltose is a disaccharide which is composed of two α–D–glucose units.

[NCERT Ed. 2025-26, 12th part II, Page No. 282]

31. (3)



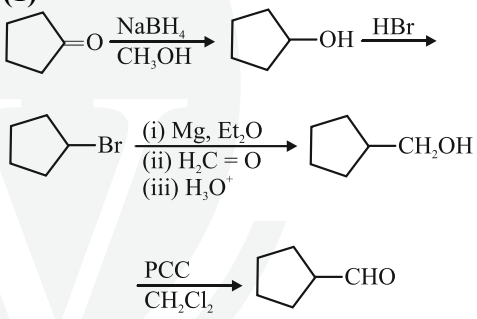
[NCERT Ed. 2025-26, 12th part II, Page No. 311]

32. (3)

Kohlrausch's law is used because molar conductivity of weak electrolytes increases on dilution and cannot be measured directly at infinite dilution. Conductivity decreases due to fewer ions, but molar conductivity increases due to reduced interionic attraction and increased dissociation.

[NCERT Ed. 2025-26, 12th part I, Page No. 50]

33. (1)



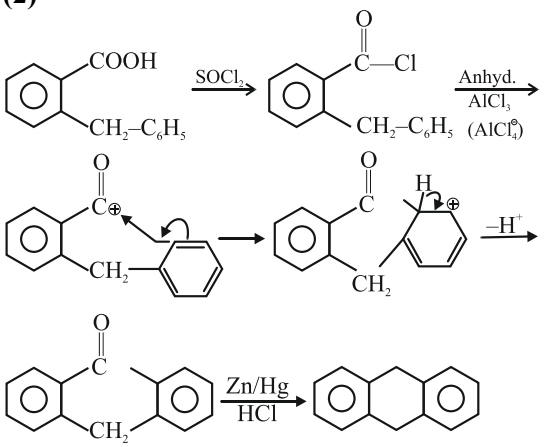
[NCERT Ed. 2025-26, 12th part II, Page No. 238]

34. (2)

Phenol give 2, 4, 6-tri-nitrophenol when it will react with conc. HNO_3 in the presence of H_2SO_4. As, –OH group is ortho-para directing. So, NO_2 occupies all the ortho and para position.

[NCERT Ed. 2025-26, 12th part II, Page No. 211]

35. (2)



[NCERT Ed. 2025-26, 12th part II, Page No. 311]

[Contd...

36. (1)

Amount of A left in n_1 halves = $\left(\frac{1}{2}\right)^{n_1}[A_0]$

Amount of B left in n_2 halves = $\left(\frac{1}{2}\right)^{n_2}[B_0]$

At the end, according to the question:

$$\frac{[A_0]}{2^{n_1}}=\frac{[B_0]}{2^{n_2}}\Rightarrow\frac{4}{2^{n_1}}=\frac{1}{2^{n_2}},[[A_0]=4[B_0]]$$

$$\therefore\ \frac{2^{n_1}}{2^{n_2}}=4\Rightarrow 2^{n_1-n_2}=(2)^2\therefore n_1-n_2=2$$

$$\therefore\ n_2=(n_1-2) \qquad \ldots(i)$$

Also $t = n_1 \times t_{1/2(A)}$ & $t = n_2 \times t_{1/2(B)}$

(Let concentration of both become equal after time t)

$$\therefore\ \frac{n_1\times t_{1/2(A)}}{n_2\times t_{1/2(B)}}=1\Rightarrow\frac{n_1\times 5}{n_2\times 15}=1\Rightarrow\frac{n_1}{n_2}=3 \qquad \ldots(ii)$$

From eq (i) and (ii),

$n_1 = 3, n_2 = 3$ (1 and 2 in subscript)

Therefore, t = 3 × 5 = 15 minute

[NCERT Ed. 2025-26, 12th part I, Page No. 72]

37. (4)

(a) Lysine (II) Basic amino acid
(b) Aspartic acid (IV) Acidic amino acid
(c) Valine (I) Neutral optically active amino acid
(d) Glycine (III) Neutral optically inactive amino acid

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38. (1)

Statement I is correct and statement-II is incorrect. Condensation of diazonium salt with phenol is carried out in weakly alkaline medium (*pH*=9). This is due to the fact that in strongly acidic medium, the $-OH$ of phenol remains unionised, and an amine forms a salt. Phenol exists as phenoxide ion and latter is readily substituted by electrophiles than phenol itself. Thus, in phenol, coupling is carried out in alkaline medium.

[NCERT Ed. 2025-26, 12th part II, Page No. 274]

39. (3)

$$\underset{(\text{Excess})}{CH_3CH_2OH}\xrightarrow[413K]{H_2SO_4}C_2H_5OC_2H_5$$

[NCERT Ed. 2025-26, 12th part II, Page No. 208]

40. (4)

Rate of solvolysis ∝ stability of carbocation

Carbocation formed in option R is least stable whereas, formed in option P is most stable.

[NCERT Ed. 2025-26, 12th part II, Page No. 169]

41. (3)

$$k_1=Ae^{-Ea_1/RT},k_2=Ae^{-E_{a_2}/RT}$$

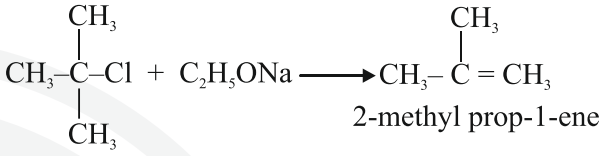
$$\frac{k_1}{k_2}=e^{\left(-E_{a_1}+E_{a_2}\right)/RT}=e^{-\left[\left(2\times10^3\right)/(2\times400)\right]}$$

$$\frac{k_1}{k_2}=e^{-2.5}=\frac{r_1}{r_2}\Rightarrow r_2=e^{2.5}r_1$$

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42. (1)

Sodium ethoxide causes dehydro halogenation of 3° alkyl and give alkene as sodium ethoxide is a strong base.



tert-butyl chloride — 2-methyl prop-1-ene

This is the limitation of williamson's synthesis

[NCERT Ed. 2025-26, 12th part II, Page No. 215]

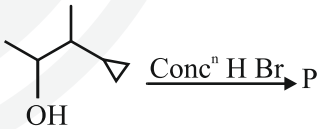
43. (2)

Aldehydes give positive tollens test, Hemi-acetals also give positive tollen's test.

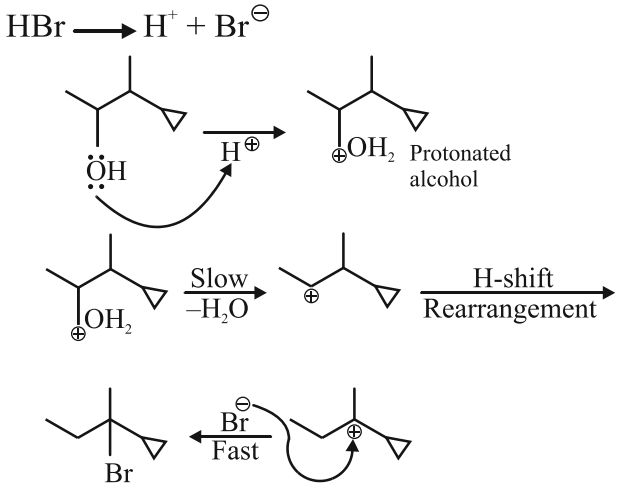
[NCERT Ed. 2025-26, 12th part II, Page No. 239]

44. (3)

It is a S_N1 reaction, so rearrangement takes place in carbocation



Mechanism:- The given alcohol is a secondary alcohol, therefore HBr will react here without catalyst. The reaction will proceed via S_N1 mechanism.



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[Contd...

45. (1)

$$E_{cell} = E^{o}_{cell} - \frac{0.059}{n}\log\frac{\left[Cr^{3+}\right]^2\left[H_2O\right]^7}{\left[Cr_2O_7^{2-}\right]\left[H^+\right]^{14}}$$

$$1.06 = 1.33 - \frac{0.059}{6}\log\frac{\left(15\times10^{-3}\right)^2(1)^7}{\left(4.5\times10^{-3}\right)\left[H^+\right]^{14}}$$

$$0.27 = \frac{0.059}{6}\log\left|\frac{225\times10^{-6}}{\left(4.5\times10^{-3}\right)\left[H^+\right]^{14}}\right|$$

$$27.45 = \log\left|\frac{50\times10^{-3}}{\left[H^+\right]^{14}}\right|$$

$$27.45 = \log\left[50\times10^{-3}\right] - \log\left[H^+\right]^{14}$$

$$27.45 = -1.30310 - 14\log\left|H^+\right|$$

$$27.45 + 1.3010 = -14\log\left|H^+\right|$$

$$28.751 = 14\times -\log\left|H^+\right|$$

$$28.041 = 14\text{pH}$$

pH = 2

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46. (3)

Acetophenone is $CH_3-\overset{O}{\overset{\|}{C}}-Ph$. It has α – H so undergoes Aldol. It has $CH_3-\overset{O}{\overset{\|}{C}}-$ group so, undergoes Iodoform.

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47. (1)

$p_1 = 80$ Pa, $p_2 = 40$ Pa

$\left(t_{1/2}\right)_1 = 350\text{sec}\left(t_{1/2}\right)_2 = 175\text{sec}$

As we know the relation between half life and concentration (or pressure) is given as, $t_{1/2} \propto p^{1-n}$ where, n is the order of the reaction.

Therefore,

$$\frac{\left(t_{1/2}\right)_1}{\left(t_{1/2}\right)_2} = \left(\frac{p_2}{p_1}\right)^{n-1}$$

$$\left(\frac{350}{175}\right) = \left(\frac{40}{80}\right)^{n-1}$$

$$2 = \left(\frac{1}{2}\right)^{n-1}$$

$$\left(\frac{1}{2}\right)^{-1} = \left(\frac{1}{2}\right)^{n-1}$$

$n - 1 = -1$

$n = -1 + 1 = 0$

n = zero i.e., zero order reaction

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48. (4)

Benzene diazonium fluroborate is water insoluble and is stable at room temperature.

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49. (1)

(a) Benzophenone Diphenylmethane

Requires complete reduction of the carbonyl group to -CH_2- Clemmensen reduction

⇒ (iii) Zn(Hg)/conc. HCl

(b) Benzaldehyde- 1-Phenylethanol

Addition of a methyl group to aldehyde followed by hydrolysis Grignard reaction

⇒ (iv) CH_3MgBr

(c) Cyclohexanone Cyclohexanol

Reduction of ketone to secondary alcohol Hydride reduction

⇒ (i) $LiAlH_4$

(d) Phenyl benzoate - Benzaldehyde

Controlled reduction of ester to aldehyde Partial reduction

(ii) DIBAL-H

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50. (3)

In Finkelstein reaction, alkyl iodides are often prepared by the reaction of alkyl chlorides/bromides with Nal in dry acetone.

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[Contd...

51. (1)

In the corn cob – the ears are stigma and style which wave in the wind to trap pollen grains.

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52. (4)

Structural gene	Transcribes *lac* mRNA
Regulator gene	Controls the activity of the operator gene
Promoter	Provides attachment site for RNA polymerase
Operator	Interacts with repressor protein

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53. (3)

Klinefelter's syndrome has the karyotype 47, XXY, which is a trisomy of sex chromosomes.

Turner's syndrome is monosomy (45, XO).

Down's syndrome is trisomy of an autosome (chromosome 21).

Myotonic dystrophy is an autosomal dominant disorder, not a chromosomal trisomy.

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54. (2)

During growth, the LAB produce acids that coagulate and partially digest the milk proteins. A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B_{12}. In our stomach too, the LAB play very beneficial role in checking disease-causing microbes.

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55. (3)

The DNA fingerprinting technique, includes steps in following sequence:

(i) isolation of DNA

(ii) digestion of DNA by restriction endonucleases,

(iii) separation of DNA fragments by electrophoresis

(iv) transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon

(v) hybridisation using labelled VNTR probe, and

(vi) detection of hybridised DNA fragments by autoradiography

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56. (2)

A is kinetochore protein. Every chromosome (visible only in dividing cells) essentially has a primary constriction or the centromere on the sides of which disc shaped structures called kinetochores are present.

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57. (2)

Histones are rich in the basic amino acid residues lysine and arginine.

(NCERT Ed. 2025-26, 12th, Page No. 83)

58. (3)

Unlike photophosphorylation where it is the light energy that is utilised for the production of proton gradient required for phosphorylation, in respiration it is the energy of oxidation-reduction utilised for the same process. It is for this reason that the process is called oxidative phosphorylation.

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59. (3)

Grasshoppers have an XO sex-determination system:

Males are XO, females are XX.

During meiosis in males, half of the sperm receive the X chromosome plus autosomes, and half receive only autosomes (no sex chromosome, O).

This produces the 1:1 ratio of X-bearing and O-bearing sperm.

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60. (3)

A key belief of the organic farmer is that biodiversity furthers health. The more variety a landscape has, the more sustainable it is. The organic farmer, therefore, works to create a system where the insects that are sometimes called pests are not eradicated, but instead are kept at manageable levels by a complex system of checks and balances within a living and vibrant ecosystem.

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61. (4)

The DNA-dependent DNA polymerases catalyse polymerisation only in one direction, that is 5'→3'. This creates some additional complications at the replicating fork. Consequently, on one strand (the template with polarity 3'→5'), the replication is continuous, while on the other (the template with polarity 5'→3'), it is discontinuous.

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62. (2)

Monocot stem lack phloem parenchyma.

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[Contd...

63. (3)

Morgan and his group found that even when genes were grouped on the same chromosome, some genes were very tightly linked (showed very low recombination) (while others were loosely linked (showed higher recombination). For example he found that the genes for eye colour and body colour were very tightly linked and showed only 1.3 per cent recombination while gene for eye colour and wing type showed 37.2 per cent recombination.

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64. (4)

Ethylene also promotes root growth and root hair formation, thus helping the plants to increase their absorption surface.

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65. (3)

In a typical nucleus, some region of chromatin are loosely packed (and stains light) and are referred to as euchromatin. The chromatin that is more densely packed and stains dark are called as Heterochromatin. Euchromatin is said to be transcriptionally active chromatin, whereas heterochromatin is inactive.

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66. (2)

Aspergillus niger	Citric acid
Propionibacterium sharmanii	Swiss cheese
Monascus purpureus	Statins
Trichoderma polysporum	Cyclosporin A

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67. (1)

In α Thalassemia, production of α globin chain is affected while in β Thalassemia, production of β globin chain is affected. α Thalassemia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16 of each parent and it is observed due to mutation or deletion of one or more of the four genes. The more genes affected, the less alpha globin molecules produced. While β Thalassemia is controlled by a single gene HBB on chromosome 11 of each parent and occurs due to mutation of one or both the genes.

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68. (1)

Anaphase I (Meiosis I):

- Homologous chromosomes separate and move to opposite poles.
- Sister chromatids remain attached at the centromere.

Anaphase II (Meiosis II):

- Centromeres split, and sister chromatids separate.

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69. (3)

A molecule that can act as a genetic material must fulfill the following criteria: (i) It should be able to generate its replica (Replication). (ii) It should be stable chemically and structurally. (iii) It should provide the scope for slow changes (mutation) that are required for evolution. (iv) It should be able to express itself in the form of 'Mendelian Characters'.

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70. (1)

Mendel conducted such artificial pollination/cross pollination experiments using several true-breeding pea lines. A true- breeding line is one that, having undergone continuous self-pollination, shows the stable trait inheritance and expression for several generations. Mendel selected 14 true-breeding pea plant varieties, as pairs which were similar except for one character with contrasting traits.

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71. (3)

As the polymorphisms are inheritable from parents to children, DNA fingerprinting is the basis of paternity testing, in case of disputes.

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72. (2)

In phenylketonuria, phenylalanine accumulates and tyrosine decreases. The disease is caused by lack of phenylalanine hydroxylase. It is an autosomal recessive, not sex-linked. It is a single-gene (Mendelian) disorder, not chromosomal.

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73. (3)

Sol. Red and black markings – the Ladybird, and Dragonflies are useful to get rid of aphids and mosquitoes, respectively.

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74. (1)

In the five kingdom classification of Whittaker there is no mention of lichens and some acellular organisms like viruses, viroids and prions. Viruses did not find a place in classification since they are not considered truly 'living', if we understand living as those organisms that have a cell structure. The viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell. Once they infect a cell, they take over the machinery of the host cell to replicate themselves, killing the host.

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75. (4)

Fucus is a brown alga (Phaeophyceae), not a bryophyte.

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[Contd...

76. (3)

The hormones which interact with intracellular receptors (mostly nuclear) like steroid hormones and iodothyronines (thyroid hormones) regulate gene expression or chromosome function.

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77. (2)

The axon terminals end in bulb-like structures called synaptic knobs, which house synaptic vesicles. These vesicles store neurotransmitters (like acetylcholine) that are released into the synaptic cleft.

Axons conduct nerve impulses away from the cell body (efferent conduction) to a synapse or a neuromuscular junction. It is the dendrites that conduct impulses toward the cell body.

Multipolar neurons have one axon and two or more dendrites and are commonly found in the cerebral cortex.

In myelinated fibres, the myelin sheath is not continuous; the gaps where the axonal membrane is exposed are called the Nodes of Ranvier, which facilitate saltatory conduction.

Unmyelinated nerve fibres in the peripheral nervous system are still enclosed by Schwann cells. However, in these fibres, the Schwann cells do not wrap repeatedly around the axon to form a thick myelin sheath. Therefore, they possess Schwann cells but lack the myelin sheath.

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78. (1)

Colostrum is rich in IgA antibodies. IgA provides local (mucosal) immunity to the infant, especially in the gut.

The foetus receives maternal IgG antibodies through the placenta, which is an example of naturally acquired passive immunity.

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79. (2)

Saheli (centchroman/ormeloxifene) is a once-a-week, non-steroidal oral contraceptive and is known for minimal side effects.

It was developed in India by the Central Drug Research Institute (CDRI), Lucknow.

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80. (4)

List-I	List-II
350 mya	Fish with stout/strong fins (Lobefins)
320 mya	Sea weeds and few plants existed
65 mya	Dinosaurs suddenly disappeared
2000 mya	First cellular forms appeared

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81. (3)

During malaria infection, *Plasmodium* multiplies inside red blood cells (RBCs).

At regular intervals (every 48 or 72 hours, depending on species), RBCs rupture.

This rupture releases haemozoin (malarial pigment) and toxins into the bloodstream.

The released substances stimulate the immune response, causing recurring chills and high fever.

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82. (3)

Sterilization methods are generally advised as a terminal method to prevent further pregnancies, they are highly effective but their reversibility is very poor, and the surgical intervention involved blocks gamete transport.

Vasectomy involves the vas deferens (male), not the fallopian tube (female; that's tubectomy).

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83. (3)

The sperm head contains an elongated haploid nucleus, and its anterior portion is covered by a cap-like structure called the acrosome. Mitochondria provide energy for the tail's movement.

Acrosomal enzymes help in fertilization.

At least 60 percent must have normal shape and size, while 40 percent must show vigorous motility.

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84. (2)

The Big Bang theory explains the origin of the universe, proposing that all matter and energy were once concentrated in a singularity that underwent a huge explosion, leading to the continuous expansion of the universe.

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[Contd...

85. (2)

The hyoid bone is a single, U-shaped bone located at the base of the buccal cavity. It is not classified as one of the 14 facial bones. The skull is typically divided into 8 cranial bones and 14 facial bones; the hyoid is a separate bone associated with the skull.

The sliding filament theory, proposed by Andrew Huxley and Hugh Huxley, is the widely accepted mechanism for muscle contraction. It states that contraction occurs when the thin filaments (actin) slide over the thick filaments (myosin), which pulls the Z-lines closer together and shortens the sarcomere without the filaments themselves changing length.

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86. (4)

In Zygote Intrafallopian Transfer (ZIFT), fertilization occurs outside the body and the zygote or early embryo (≤8 cells) is placed into the fallopian tube.

In Gamete Intrafallopian Transfer (GIFT), gametes (sperm and ovum) are transferred into the fallopian tube.

Intracytoplasmic Sperm Injection (ICSI) involves injecting a single sperm directly into the cytoplasm of an egg.

Intrauterine Insemination (IUI) places processed semen directly into the uterus to increase chances of fertilization.

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87. (2)

Homo habilis had a brain capacity of 650-800cc, *Homo erectus* had a brain capacity of around 900cc, and Neanderthal man had a brain capacity of 1400cc.

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88. (2)

Viral RNA is converted into viral DNA by reverse transcriptase.

The virus first enters macrophages and then T-lymphocytes.

Macrophages act as an 'HIV factory', continuously producing virus particles and progressive decrease in helper T-lymphocytes .

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89. (2)

Amniocentesis is a procedure used for testing for genetic disorders, but is currently under a statutory ban for sex determination to prevent female foeticide.

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90. (3)

Von Baer disapproved Haeckel's proposal by noting that embryos never pass through the adult stages of other animals, contradicting the idea that embryonic stages represent adult ancestral forms.

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91. (4)

List-I	List-II
Hydrolases	Enzymes catalysing hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide or P-N bonds
Lyases	Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds
Isomerases	Includes all enzymes catalysing inter-conversion of optical, geometric or positional isomers
Ligases	Enzymes catalysing the linking together of 2 compounds

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92. (3)

Malignant tumors are masses of neoplastic cells that grow rapidly, invade tissues, and exhibit metastasis—the property where sloughed cells reach distant sites to start new tumors.

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93. (1)

The diagram shows disruptive (diversifying) selection — individuals at both extremes of a trait are favored, while the intermediate phenotype is selected against, producing two peaks in the population.

Snails with very light and very dark shells are favored; medium-colored shells are preyed upon → This is classic disruptive selection (both extremes favored).

Intermediate human birth weight has lowest mortality → Stabilizing selection (average favored).

Peppered moths becoming darker over time → Directional selection (one extreme favored).

Only deer with average running speed survive → Stabilizing selection (extremes removed).

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[Contd...

94. **(1)**

List-I	List-II
IUDs	Increase phagocytosis of sperm within uterus
Barrier (Diaphragm)	Prevents conception by blocking entry of sperm through cervix
Natural (Withdrawal)	Male withdraws penis just before ejaculation
Surgical	Blocks gamete transport

(NCERT Ed. 2025-26, 12th, Page No. 44)

95. **(1)**

In the human kidney, the osmolarity increases from the outer cortex (~300 mOsmol/L) to the inner medulla (~1200 mOsmol/L). This gradient is established and maintained by the counter-current mechanism (involving the Henle's loop and *vasa recta*).

The arrangement of Henle's loop and *vasa recta* is responsible for maintaining the gradient, the specific solutes involved are NaCl and Urea. Ammonia is secreted into the tubules to maintain pH and ionic balance, but it is not the solute recycled to maintain the hyperosmolarity of the medullary interstitium. The gradient is mainly caused by NaCl and urea.

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96. **(3)**

An ECG records the electrical activity of the entire heart, including both the atria (P-wave) and the ventricles (QRS complex and T-wave). It is used to study a wide range of cardiac disorders.

While the P-wave reflects atrial activity, the heart rate of an individual is clinically determined by counting the number of QRS complexes (or R-waves) per minute, as these represent the ventricular contractions that produce a pulse.

A standard ECG has a specific shape and timing for each wave (P, QRS, and T). Any deviation from this normal pattern—such as an elevated ST segment, a prolonged PR interval, or an inverted T-wave—is a clinical indicator of potential cardiac abnormalities or diseases (e.g., heart block, arrhythmia, or myocardial infarction).

The T-wave represents ventricular repolarization (the return of the ventricles to a resting state), which marks the end of ventricular systole. The beginning of ventricular systole is represented by the QRS complex.

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97. **(2)**

Adaptive radiation involves diversification of multiple species from a common ancestral stock (as in Darwin's finches and Australian marsupials), whereas the evolution of the eye in octopus and mammals is convergent evolution.

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98. **(4)**

Medical Termination of Pregnancy (MTP) is relatively safe during the first trimester. Risk increases in the second trimester.

In India, MTP was legalized in 1971.

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99. **(3)**

Heroin (diacetylmorphine) is a depressant.

Cannabinoids are obtained from the inflorescences of *Cannabis sativa*.

Cocaine is obtained from *Erythroxylum coca* and blocks dopamine reuptake, interfering with dopamine transport.

Morphine is extracted from the latex of *Papaver somniferum* (opium poppy).

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100. **(2)**

Graft rejection mainly occurs due to the cell-mediated immune response.

T-lymphocytes recognize the transplanted organ as non-self and attack it.

Cytotoxic T cells are especially responsible for destroying graft cells.

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